Flagler Health

—

Predicting Patient Drop-off

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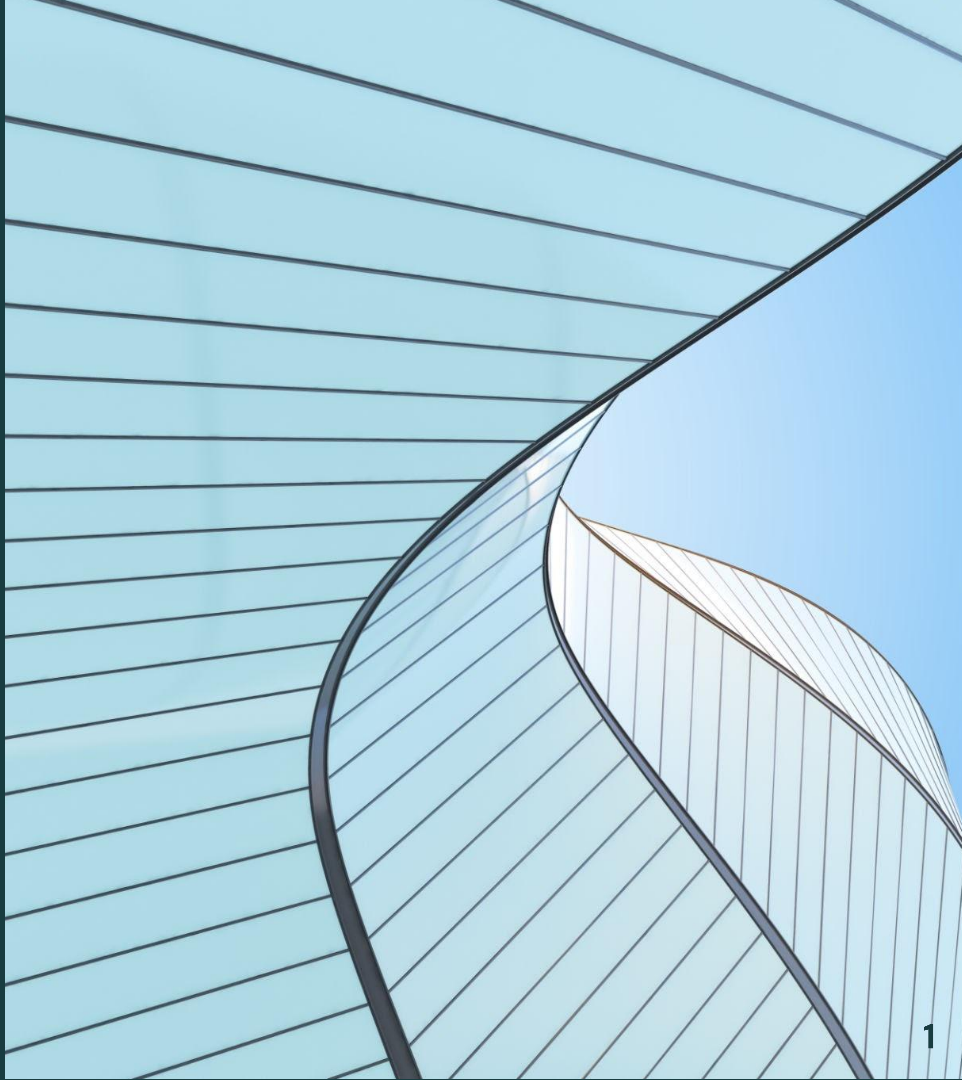

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Project Overview

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- Flagler Health provides *Remote Therapeutic Monitoring* (RTM) services for patients with musculoskeletal conditions (e.g., back pain, knee pain)

RTM (Remote Therapeutic Monitoring)

- Program from Centers for Medicare & Medicaid Services (CMS)
- Tracks non-physiological data
    - e.g. patient behavior, adherence to therapy
- Supports insurance reimbursement for ongoing patient monitoring

Project Overview

Why this matters

- Revenue + care quality depend on continuous patient engagement
- Patients often enroll → disengage/drop out
- Dropout → ↓ treatment effectiveness + ↓ billable monitoring

Goal

- Analyze patient data + Medical Assistant interactions
- Identify patterns in engagement, adherence, dropout
- Inform strategies to improve retention + outcomes

Project Overview

Research Question:

<u>What factors</u> can we use to predict <u>patient disengagement</u>?

- Baseline Metrics
- Claims
- Text

Patients who enrolled in RTM but stop actively participating over time

(e.g., reduced messaging, incomplete therapy tracking)

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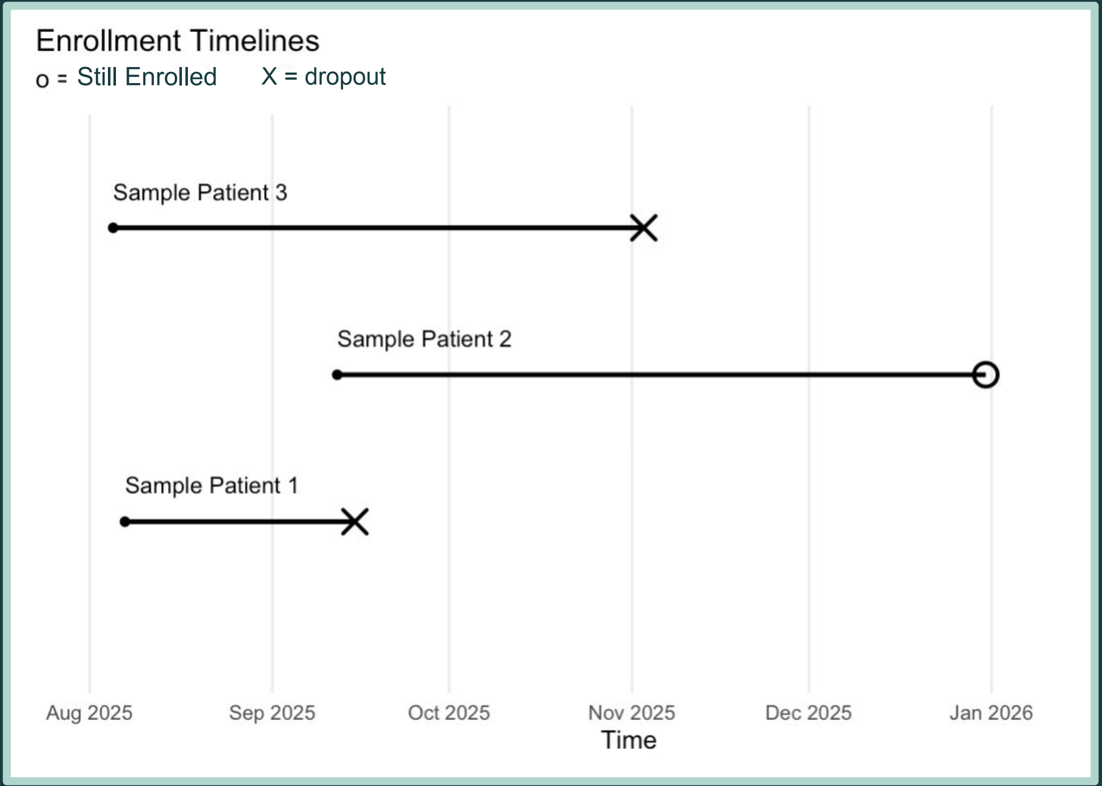
Data Overview

Data Overview

Collected by Medical Assistants who obtain patient metrics via phone check-ins:

- 391 patients
- Time window ends at 12/31 due to insurance policies

Examine patients at **baseline** for time to drop out.



Data Overview

Metrics

- **Numeric Pain Rating Scale** (NPRS) [0-10]
  - Modified to 4 categories based on quartiles
- **Mobility level** (Not Active / Active / Very Active)
- **Mood** (Sad / Flat / Energetic)
- **Sleep** (Poor / Moderate / Restful)

0%	25%	50%	75%	100%
0	4	6	7	10

Baseline calculated individually for each metric by taking first entry after enrollment date

Data Overview

Status Change

- Main categorical outcome variable
  - Added a dropped_out column
    - 1 = Dropped Out
    - 0 = Still Enrolled
- Kept Enrollment and Disenrollment date to calculate days until last observation


```
[
  {
    "chat": "68a3a09ce646deef537bb86b",
    "text": "Set RTM status to DECLINED",
    "ts": "2025-09-03T19:53:34.100Z"
  },
  {
    "chat": "68a3a09ce646deef537bb86b",
    "text": "Set RTM status to ENROLLED",
    "ts": "2025-08-26T21:35:57.842Z"
  }
]
```

Data Overview

Insurance (Medical Billing History)

- **Procedure Codes**:
  - Ex: 99202: a new patient visit involving a straightforward medical decision-making process for minor problems
- **Diagnosis Codes**:
  - Ex. M47812: Spondylosis without myelopathy or radiculopathy, cervical region
  - Translate: Age-related neck arthritis without severe nerve or spinal cord damage.

Key Category (character)	CPT Code Range(s) (character)
Evaluation & Management (E/M)	99202-99499
Anesthesia	00100-01999, 99100-99140
Surgery	10004-69990
Radiology	70010-79999
Pathology & Laboratory	80047-89398
Medicine	90281-99199, 99500-99607

Baseline calculated by taking closest entry to enrollment date (before or after)

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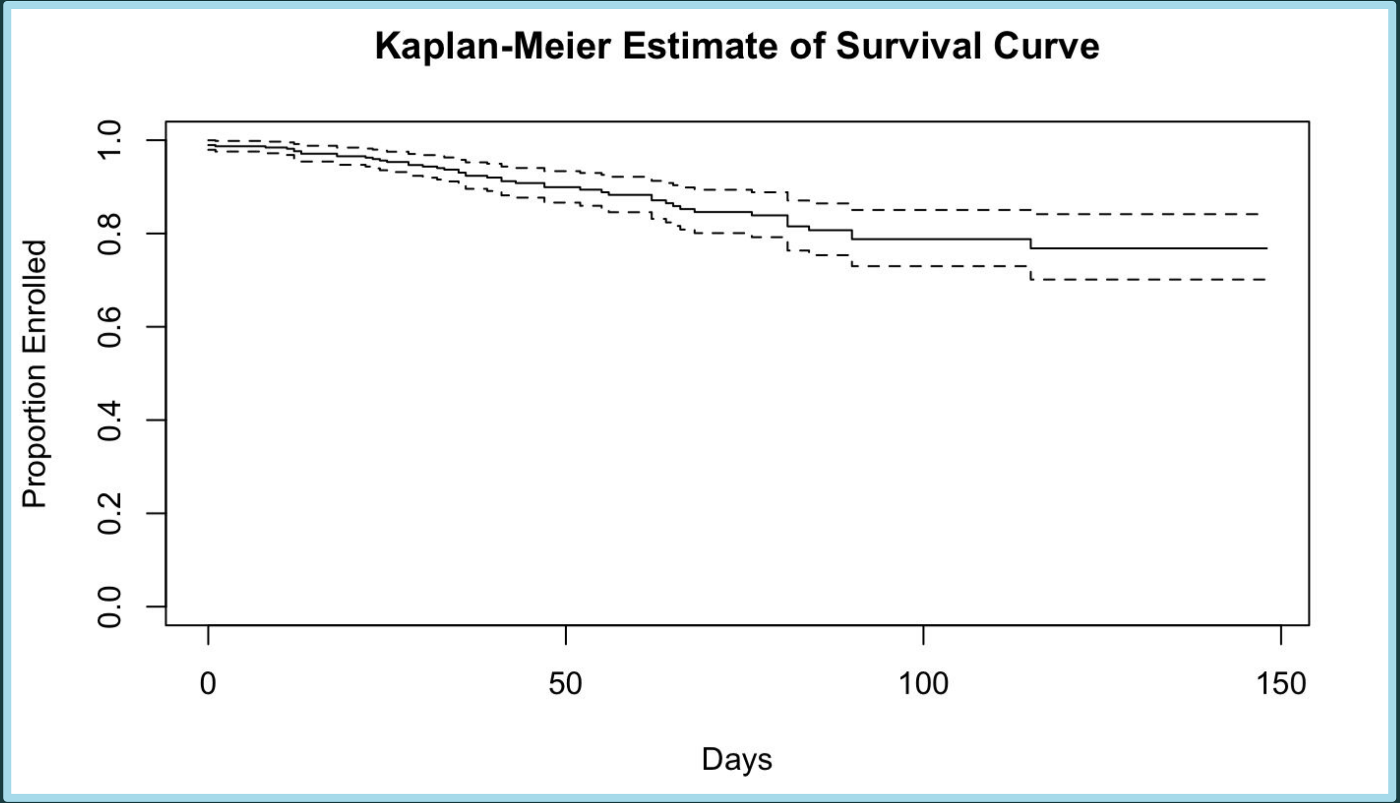
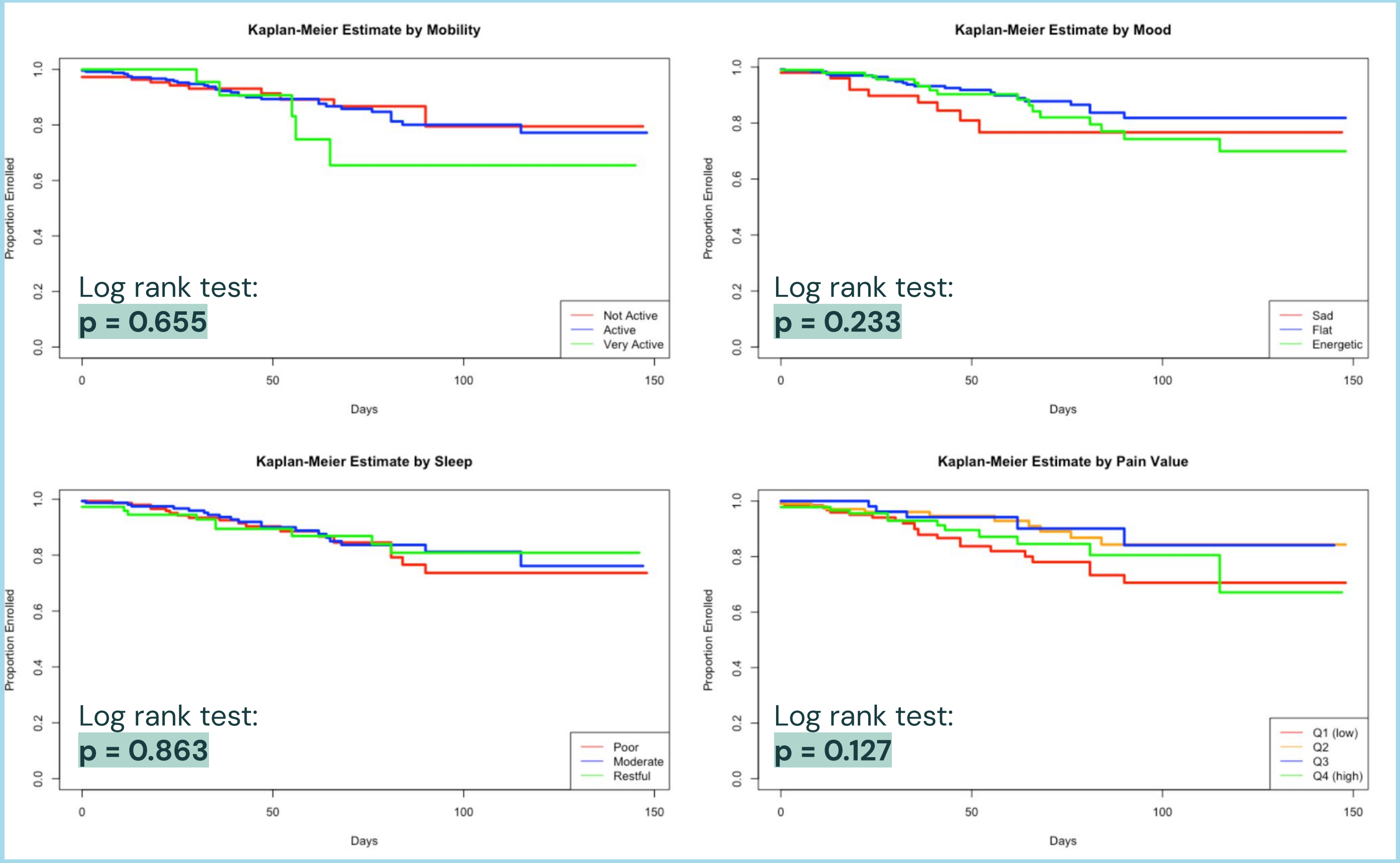
Time to Dropout

Time To Dropout By Survival Analysis



Main Kaplan-Meier Curve for the overall time-event

More Survival Analysis



More Survival Analysis

```
Call:
coxph(formula = Surv(days_enrolled, is_enrolled == 0) ~ pain_centered +
    pain_centered_squared + mobility + mood + sleep, data = patients_complete)

  n= 383, number of events= 49

                           coef exp(coef)  se(coef)      z Pr(>|z|)
pain_centered         -0.102457  0.902617  0.072872 -1.406   0.1597
pain_centered_squared -0.005044  0.994968  0.019967 -0.253   0.8006
mobilityMedium         0.169051  1.184181  0.363302  0.465   0.6417
mobilityHigh           0.500256  1.649143  0.565326  0.885   0.3762
moodMedium            -0.692305  0.500421  0.418277 -1.655   0.0979 .
moodHigh              -0.440499  0.643715  0.471385 -0.934   0.3501
sleepMedium           -0.264765  0.767386  0.338852 -0.781   0.4346
sleepHigh             -0.252935  0.776518  0.414302 -0.611   0.5415
```

- No predictors are statistically significant at the 0.05 level
- Pain was mean-centered to reduce potential multicollinearity
- Model Extension: Included a squared pain term to capture potential non-linear (polynomial) relationships

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Claims: Procedure and Diagnosis

Procedure

Table 1: Summary of Procedures

Procedure Type	Count
Evaluation and Management	203
Medicine	44
Pathology and Laboratory	8
Radiology	8
Surgery	127

→

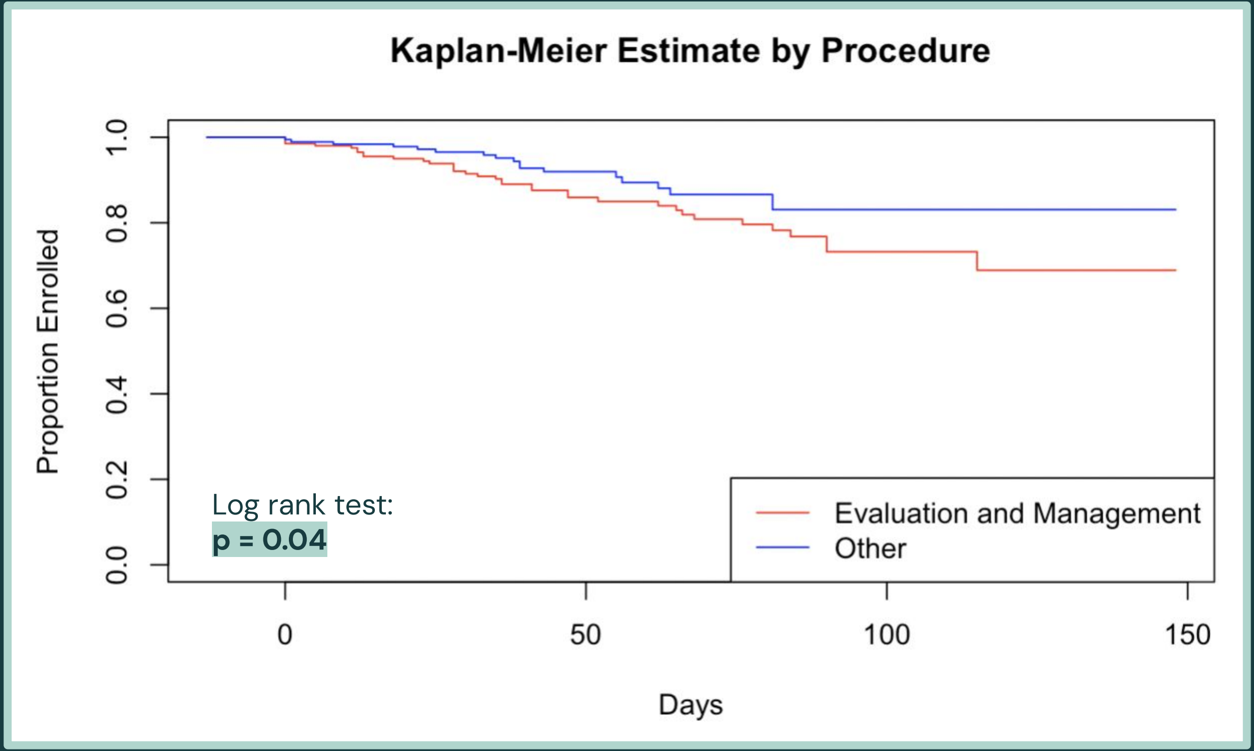
Table 2: Summary of Procedures (2 Groups)

Procedure Type	Count
Evaluation and Management	203
Other	187

- Balanced comparison

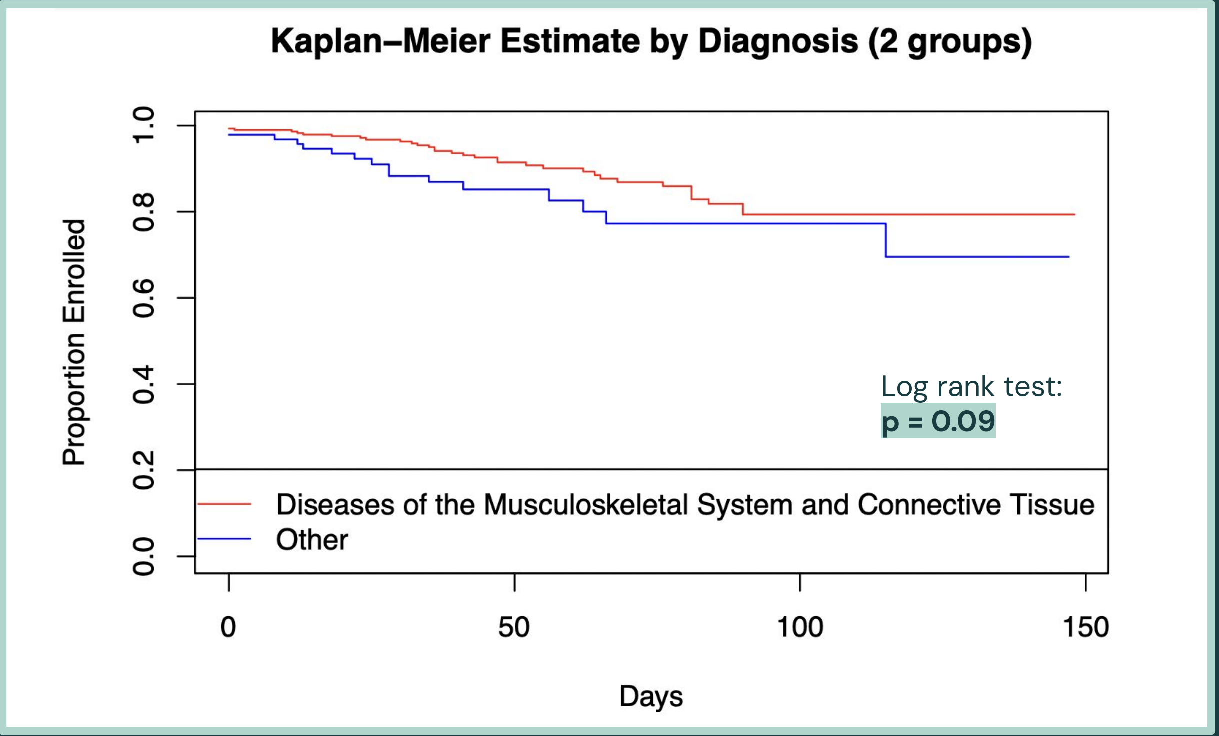
- Distribution highly skewed
- Insufficient sample sizes

Time to Dropout



Diagnosis

- Categories: Musculoskeletal and Other



Model

1. **Cox Proportional Hazards:** Analyzes the time until a patient drops out
2. **Logistic Regression:** Analyzes whether a patient drops out or not

Comparison

- Similar → strengthens our confidence in the findings
- Different → reveal whether timing plays an important role

Variables

- `Drop out ~ Pain + Mobility + Mood + Sleep +`
  `Procedure Group + Diagnosis Group`

Model Results

Cox Proportional Hazards

Baseline Pain

HR: 0.90

10% lower dropout risk per unit increase in pain score

$p = 0.07$

Procedure Group

HR: 0.56

"Other" procedures show a 44% lower dropout risk vs. Evaluation & Management

$p = 0.06$

Diagnosis Group

HR: 1.70

"Other" diagnoses have a 70% higher dropout risk vs. Musculoskeletal

$p = 0.09$

- Baseline metrics: $p > 0.1$
- **Logistic regression:** coefficients align closely with the Cox model results

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Text: Sentiment Analysis

Messages

- Longitudinal text history stored as JSON files

- Interactions typically initiated by medical assistants (MAs) to monitor patient metrics and provide follow-up support

- **Preprocessing**: Grouped messages by patient_id and date


```
{
  "chat_id": "68a4a08b43406f2b699bc48d",
  "type": "voice",
  "sender": "66d5fdb356c6577749cae7c4",
  "role": "Care Counselor",
  "ts": "2025-10-22T17:27:29.199Z",
  "phonecall_id": "68f913b20dd0b0413e5d2717",
  "text": "I'd like to give you more information about the remote therapeutic monitoring program. We'll be tracking your pain lev
},
{
  "chat_id": "68a4a08b43406f2b699bc48d",
  "type": "voice",
  "sender": "682f78a7ff28a795af96024f",
  "role": "Patient",
  "ts": "2025-10-22T17:29:02.989Z",
  "phonecall_id": "68f913b20dd0b0413e5d2717",
  "text": "Wow, that sounds good. I'd love to be pain free. My back is my biggest problem."
},
```

Sentiment Analysis

1. **Sentiment Model**: Pretrained NLP model (Flair)
   - Output:
     - Label (Positive / Negative)
     - Confidence score (0–1)
2. Calculate continuous score: multiply confidence by 1 if positive, or –1 if negative

patient_id	date	full_text	sentiment	confidence
682f7806186f11(	2025-09-18	Amanda're too kind!	POSITIVE	0.9902254939
682f7806186f11(	2025-12-11	Here's some informa	NEGATIVE	0.998288095
682f77ec186f11(	2025-11-03	Hi Sarah, hope every	POSITIVE	0.991508007
682f7863186f11(	2025-09-29	Hi Kay. We're checki	NEGATIVE	0.951164186
682f7863186f11(	2025-10-09	Hi Gaby. Ashley bacl	NEGATIVE	0.6664472818

Logistic Regression

Model

```
Drop out ~ Pain + Mobility + Mood + Sleep +
    Procedure Group + Diagnosis Group +
    Sentiment
```

Result

- Sentiment:
  - Coefficient: -1.29 (p-value: 0.01) → odds ratio: exp(-1.29) ≈ 0.28
  - More positive → 72% lower odds of dropout
- Similar trends observed for other predictor variables

Model Comparison

AIC

- With Sentiment: 294.02
- Without Sentiment: 299.36

ANOVA

- p-value = 0.00673

Hosmer-Lemeshow

- Calibration Check for Logistic Regression
    - p-value = 0.72

All 3 of our model robustness tests pass!

Table: Observed vs. Expected Dropouts by Risk Decile

Group	N	Observed	Expected	Avg_p_hat	Chi_Sq
1	39	0	1.17	0.030	1.167
2	39	1	2.01	0.052	0.508
3	39	2	2.62	0.067	0.146
4	38	5	3.22	0.085	0.990
5	38	3	3.91	0.103	0.212
6	38	6	4.67	0.123	0.382
7	38	7	5.45	0.143	0.442
8	38	8	6.71	0.177	0.248
9	38	9	7.92	0.208	0.148
10	38	8	11.34	0.298	0.982

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Conclusion & Next Steps

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- ***Procedure grouping***
  - Clearest differences in dropout patterns
  - Consistent across Cox and Logistic models
  - More intense care associated with stronger patient engagement

- ***Sentiment***
  - Statistically significant predictor of dropout
  - Higher positive sentiment → lower likelihood of dropout

Conclusion & Next Steps

Next Steps:

- Further analysis of insurance categorization
- Explore data more longitudinally
  - Ex: pain level over time
- Explore chat data further
  - Response rate
  - Sentiment of last message before drop out
- Interactions between variables
  - Ex: Sentiment x Procedure Group

Thank you!